| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

$\begin{bmatrix} 4 \\ -2 \\ -2 \end{bmatrix}$ *Initial W* **REPEAT using Step η = 1**

$C_1$ positive $C_2$ negative **Zero** is unaccepted

Check classes of

| $Z_1$ | $Z_2$ | $Z_3$ |
|---|---|---|
| 0 | 7 | 3 |
| 0 | 6 | 2 |

*Ugmented Vectors (from Top)*

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 1 | 5 | 6 | 6 |
| 1 | 1 | 2 | 4 | 5 | 6 |
| $C_1$ | $C_1$ | $C_1$ | $C_2$ | $C_2$ | $C_2$ |

## Iteration 2

No change in W

*Stop* Iterations

$d(x_1) = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix} = 3$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix}$

$d(x_2) = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix}$

$d(x_3) = \begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix} = 2$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix}$

$d(x_4) = \begin{bmatrix} 1 & 5 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix} = -4$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix}$

$d(x_5) = \begin{bmatrix} 1 & 6 & 5 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix} = -6$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix}$

$d(x_6) = \begin{bmatrix} 1 & 6 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix} = -7$ ⇒ Keep same Weight $\underline{W} = \begin{bmatrix} 5 \\ -1 \\ -1 \end{bmatrix}$